

HW11

8.2 (a) $y(t) = e^{j\omega_c t} x(t) \Rightarrow x(t) = e^{-j\omega_c t} y(t)$

no constraint $B = 1000\pi$

(b) $y(t) = (\cos \omega_c t + j \sin \omega_c t) x(t)$

$$\text{Re}\{y(t)\} = \frac{1}{2} e^{j\omega_c t} x(t) + \frac{1}{2} e^{-j\omega_c t} x(t)$$

$$\text{Re}\{y(j\omega)\} = \frac{1}{2} X(j\omega - \omega_c) + \frac{1}{2} X(j\omega + \omega_c)$$

$$[\omega_c - B, \omega_c + B] \cup [-\omega_c - B, -\omega_c + B]$$

$$\omega_c - B > -\omega_c + B \Rightarrow \omega_c > B$$

or $\omega_c + B < -\omega_c - B \Rightarrow \omega_c < -B \therefore |\omega_c| > B$

$$\omega_c > 1000\pi$$

8.3 $\sin(\omega_c t) = \frac{1}{2j} (e^{j\omega_c t} - e^{-j\omega_c t})$

$$Y(t)g(t)\cos\omega_c t = \frac{1}{2} x(t)\sin 2\omega_c t$$

$$Y(j\omega) = \frac{1}{4j} [X(j(\omega - 2\omega_c)) - X(j(\omega + 2\omega_c))]$$

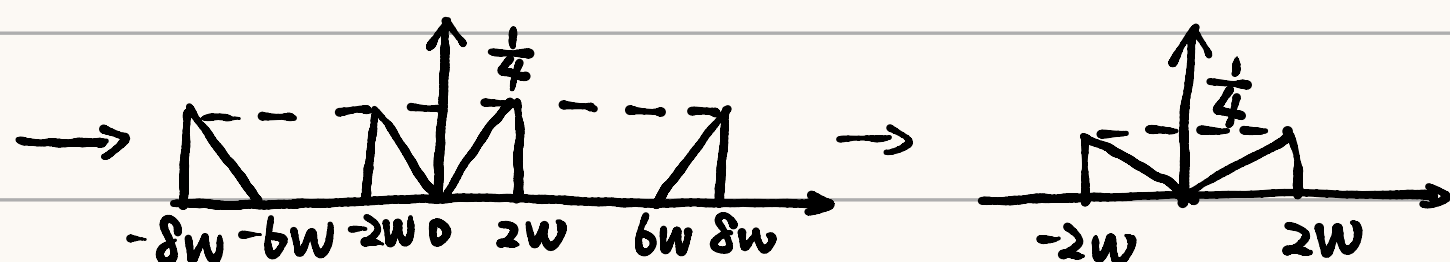
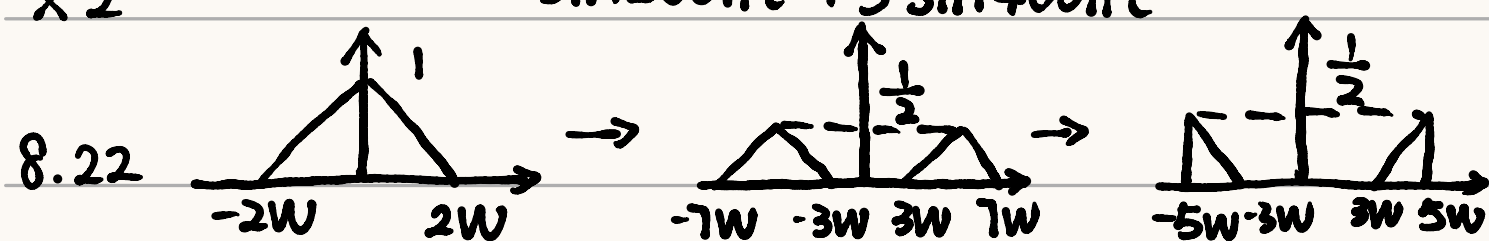
$$[-6000\pi, -2000\pi] \cup [2000\pi, 6000\pi] \therefore y(t) = 0$$

8.4 $g(t)\sin 400\pi t = x(t)\sin^2 400\pi t = \frac{1}{2}x(t) - \frac{1}{2}x(t)\cos(800\pi t)$

$$= \frac{1}{2}\sin 200\pi t + \sin 400\pi t - \frac{1}{2}\sin 200\pi t \cos 800\pi t - \sin 400\pi t \cos 800\pi t$$

after filter $= \frac{1}{2}\sin 200\pi t + \frac{3}{2}\sin 400\pi t$

$$\times 2 = \sin 200\pi t + 3\sin 400\pi t$$



8.23 $y(t) = x(t)\cos(\omega_c t)$ $w(t) = y(t)\cos\omega_d t$

$$w(t) = \frac{1}{2} x(t)\cos(\omega_c - \omega_d)t + \frac{1}{2} x(t)\cos(\omega_c + \omega_d)t$$

$$\Delta\omega = \omega_c - \omega_d$$

$\therefore$ low pass $w'(t) = \frac{1}{2} x(t)\cos(\Delta\omega t)$

$\therefore w'(t) \propto x(t)\cos(\Delta\omega t)$

